Example 2.5. Let $p \in \mathbb{Z}$ be a prime. Then $\mathbb{Z}_{(p)} = \{a/b : a, b \in \mathbb{Z}, p \nmid b\}$. As in the previous example, $\mathbb{Z}$ is a PID and $\mathbb{Z}_{(p)}$ is a DVR; the valuation on $\mathbb{Q}$ is the p-adic valuation. The residue field is $\mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)} \simeq \mathbb{F}_p$ and the quotient map $\mathbb{Z}_{(p)} \twoheadrightarrow \mathbb{F}_p$ is reduction modulo p.

2.3 Localization of modules

The concept of localization generalizes immediately to modules. As above, let A be a commutative ring, let S be a multiplicative subset of A, and let M be an A-module. The localization $S^{-1}M$ of M with respect to S is an $S^{-1}A$-module equipped with an A-module homomorphism $\iota\colon M \to S^{-1}M$ with the universal property that if N is an $S^{-1}A$-module and $\varphi\colon M \to N$ is an A-module homomorphism, then φ factors uniquely through $S^{-1}M$ (via ι). Note that in this definition we are viewing $S^{-1}A$-modules as A-modules via the canonical homomorphism $A \to S^{-1}A$ that is part of the definition of $S^{-1}A$. Our definition of $S^{-1}M$ reduces to the definition of $S^{-1}A$ in the case $M = A$.

The explicit construction of $S^{-1}M$ is exactly the same as $S^{-1}A$, one takes the quotient of $M \times S$ modulo the same equivalence relation as in (1):

$$(a, s) \sim (b, t) \Leftrightarrow \exists u \in S \text{ such that } u(ta - sb) = 0,$$

where a and b now denote elements of M, and $\iota(a) := a/1$ as before. Alternatively, one can define $S^{-1}M := M \otimes_A S^{-1}A$ (see [2, Prop. 3.5] for a proof that this is equivalent). In other words, $S^{-1}M$ is the *base change* of M from A to $S^{-1}A$; we will discuss base change more generally in later lectures.

The map $\iota\colon M \to S^{-1}M$ is injective if and only if the map $M \xrightarrow{\times s} M$ is injective for every $s \in S$. This is a strong condition that does not hold in general, even when A is an integral domain (the annihilator of M may be non-trivial), but it applies to all the cases we care about. In particular, if A is a subring of a field K (in which case A must be an integral domain whose fraction field lies in K) and M is an A-module that is contained in a K-vector space. In this setting multiplication by any nonzero $s \in A$ is injective and we can view M as an A-submodule of any of its localizations $S^{-1}M$.

We will mostly be interested in the case $S = A - \mathfrak{p}$, where $\mathfrak{p}$ is a prime ideal of A, in which case we write $M_\mathfrak{p}$ for $S^{-1}M$, just as we write $A_\mathfrak{p}$ for $S^{-1}A$.

Proposition 2.6. *Let A be a subring of a field K, and let M be an A-module contained in a K-vector space V (equivalently, for which the map $M \to M \otimes_A K$ is injective).*[1] *Then*

$$M = \bigcap_\mathfrak{m} M_\mathfrak{m} = \bigcap_\mathfrak{p} M_\mathfrak{p},$$

where $\mathfrak{m}$ ranges over the maximal ideals of A, $\mathfrak{p}$ ranges over the prime ideals of A, and the intersections take place in V.

Proof. The fact that $M \subseteq \bigcap_\mathfrak{m} M_\mathfrak{m}$ is immediate. Now suppose $x \in \bigcap_\mathfrak{m} M_\mathfrak{m}$ and consider the A-ideal $\mathfrak{a} := \{a \in A : ax \in M\}$. For each maximal ideal $\mathfrak{m}$ we can write $x = m/s$ for some $m \in M$ and $s \in A - \mathfrak{m}$; we then have $sx \in M$ and $s \in \mathfrak{a}$, but $s \notin \mathfrak{m}$, so $\mathfrak{a} \not\subseteq \mathfrak{m}$. This holds for every maximal ideal $\mathfrak{m}$, so $\mathfrak{a} = (1)$. Thus $1 \in \mathfrak{a}$ and $x = 1 \cdot x \in M$; thus $\cap_\mathfrak{m} M_\mathfrak{m} \subseteq M$.

Each $\mathfrak{p}$ lies in some $\mathfrak{m}$ for which $M_\mathfrak{m} \subseteq M_\mathfrak{p}$, and every $\mathfrak{m}$ is a $\mathfrak{p}$, so $\cap_\mathfrak{m} M_\mathfrak{m} = \cap_\mathfrak{p} M_\mathfrak{p}$. $\square$

[1] Here $M \otimes_A K$ is a tensor product of A-modules that is also a K-vector space. We need the natural map to be injective in order to embed M in it. Note that V necessarily contains a subspace isomorphic to $M \otimes_A K$.